

**CPCS-241 FINAL
(DATABASE)
2023 2nd SEMESTER**

PART I: SQL Queries

Show the output:

```
SELECT year, season, count(*)  
FROM semester s, student d  
WHERE s.studentId = d.studentId  
GROUP BY semester#
```

```
SELECT name, stuID, Max(y) FROM student  
WHERE (SELECT name, stuID, Count(*) AS y  
FROM student s, Enroll e WHERE s.stuID = e.stuID)  
GROUP BY courseN
```

Not sure about the queries but it's the same idea

STUDENT

name	stuID
Dana A.	101
Ruba H.	102
Sara T.	103
Huda W.	104
Jana S.	105
Ghala M.	106
Amal F.	107

ENROLL

CourseN	stuId
CPCS241	101
CPCS202	102
CPCS203	103
CPCS204	104
CPCS222	105
CPCS241	106

SEASON

COURSE

DEPARTMENT

SEMESTER

Q1. Choose the equivalent Relational Algebra statements for

```
SELECT C.CrsName
FROM Course C, Teaching T
WHERE C.CrsCode=T.CrsCode AND T.Semester='S2000'
```

- a) $\pi_{\text{CrName}}(\text{Course} \bowtie \sigma_{\text{semester}='S2000'}(\text{Teaching}))$
- b) $\pi_{\text{semester}='S2000'}(\text{Course} \bowtie \sigma_{\text{semester}='S2000'}(\text{Teaching}))$
- c) $\pi_{\text{CrName}}(\text{Course} \bowtie \sigma_{\text{semester}}(\text{Teaching}))$

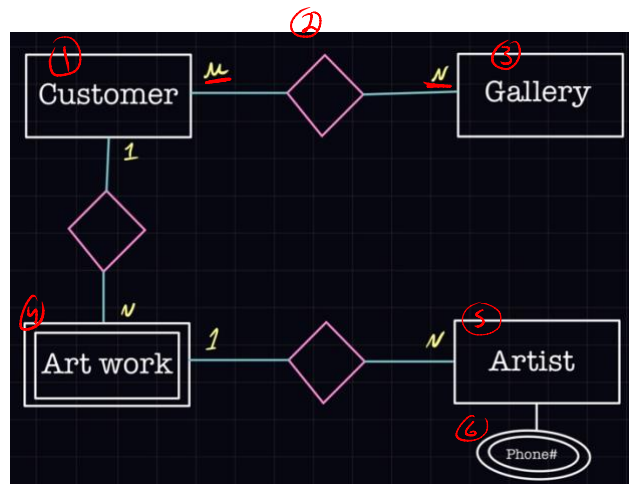
Q2. Choose the equivalent sql statements for

$$\pi_{\text{ProfId}}(\sigma_{T1.\text{CrCode} \neq T2.\text{CrCode}}(\text{Teaching}[\text{ProfId}, T1.\text{CrCode}, \text{Semester}] \bowtie \text{Teaching}[\text{ProfId}, T2.\text{CrCode}, \text{Semester}]))$$

- a) SELECT T1.ProfId
FROM Teaching T1, Teaching T2 WHERE T1.ProfId = T2.ProfId AND T1.Semester = T2.Semester AND T1.CrsCode <> T2.CrsCode
- b) SELECT T1
FROM Teaching T1, Teaching T2 WHERE T1.ProfId = T2.ProfId AND T1.Semester = T2.Semester AND T1.CrsCode <> T2.CrsCode
- c) SELECT T1.ProfId
FROM Teaching T1 WHERE T1.ProfId = T2.ProfId AND T1.Semester = T2.Semester AND T1.CrsCode <> T2.CrsCode

Q2. How many relations will be represented from the following ER ?

.....6..... Relations.



PART II: FILE ORGANIZATION AND INDEXING

Q1.

Inserting and deleting records are expensive operations for an ordered file	☒ True ☐ False
Average blocks access a specific record for ordered file is $\log_2 b$	☒ True ☐ False

Q2. Consider a disk with block size $B = 512$ bytes. Record size R in bytes 115 bytes, calculate the blocking factor bfr.

- a) 4
- b) 5
- c) 2
- d) 3

$$bfr = \lfloor B/R \rfloor = \lfloor 512/115 \rfloor = \lfloor 4.45 \rfloor = 4$$

Q3. Use Primary Index to find the cost of search, if the number of blocks occupied by BOOKSTORE file is 59, block size is 519 bytes, record length = 13.

$$b = \lfloor B/R \rfloor = \lfloor 519/13 \rfloor = 39$$

$$bfr = \lfloor B/R \rfloor = \lfloor 519/13 \rfloor = \lfloor 39.92 \rfloor = 39$$

$$bi = \lceil r/bfr \rceil = \lceil 151/39 \rceil = 4$$

Cost :-

$$\lceil \log_2(b) \rceil = \lceil \log_2(39) \rceil = 5$$

1 + 1 block Access

$$= \underline{\underline{6}}$$

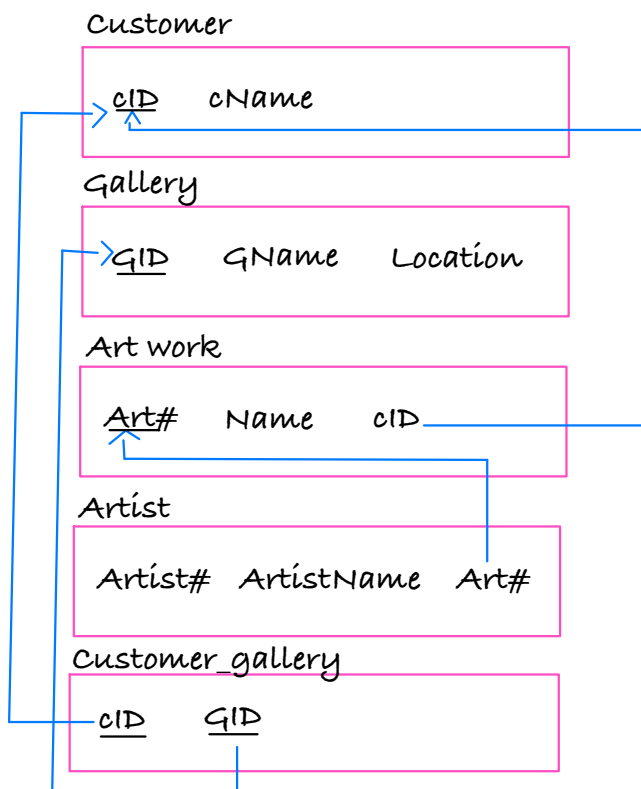
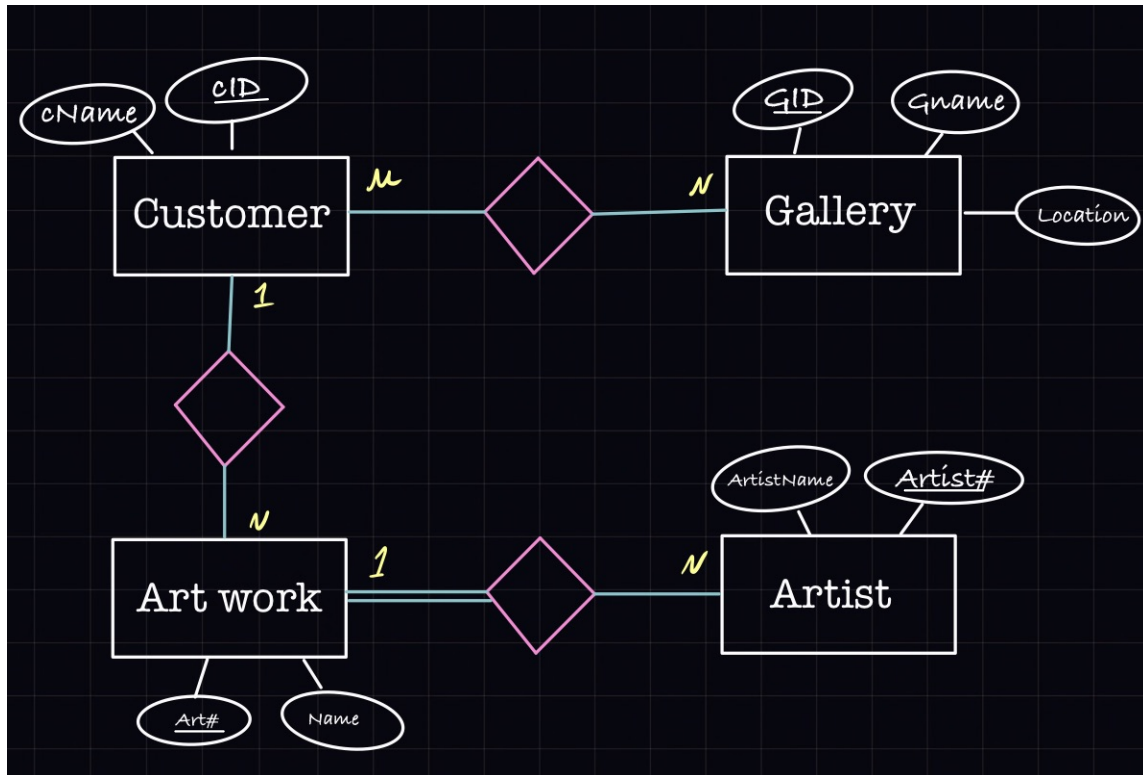
PART III: ER Diagram

ER Diagram Question, with participation and cardinality, it was about Car Reservation system,
Entities: Customer, Car, Carpark, Reservation, City, Country

The customer reserves a car and pick the car from the car park, car park located in many cities
which the city in a country.

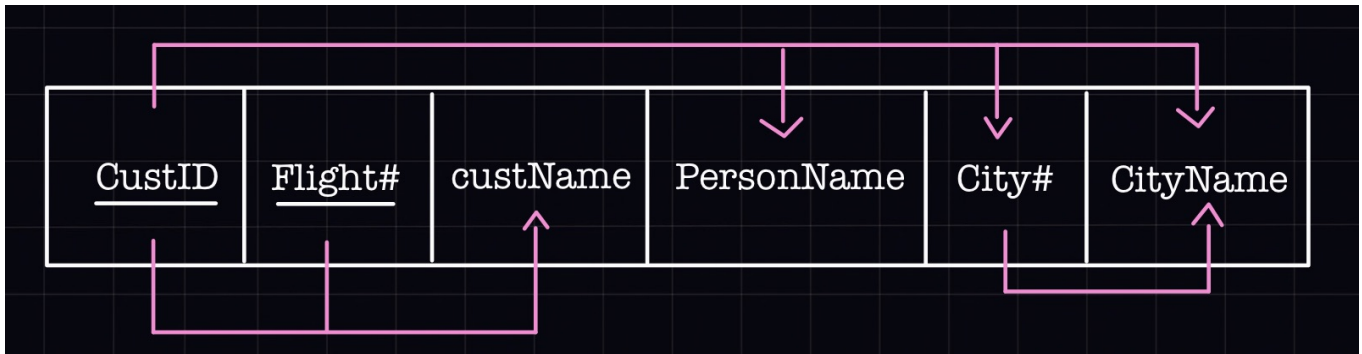
PART IV: Mapping ER to Schema

Map the following ER to Schema

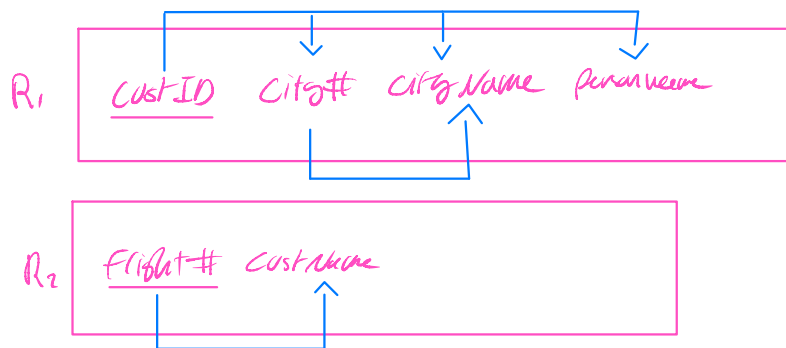


PART V: Normalization

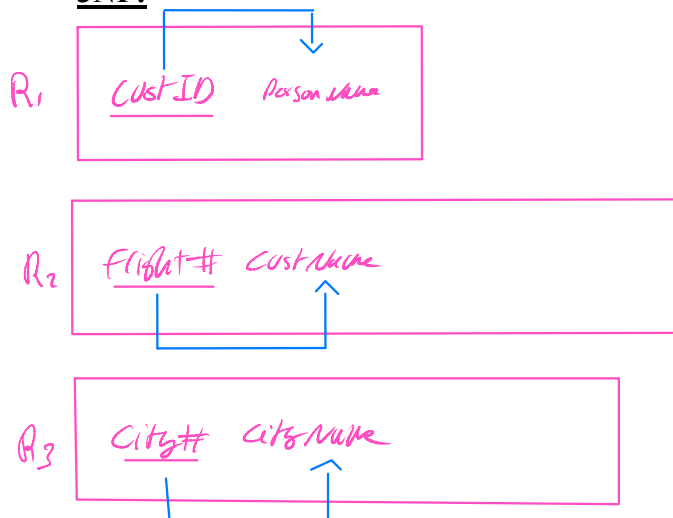
The following schema is in the First NF, normalize it to 2NF & 3NF and show the final schema



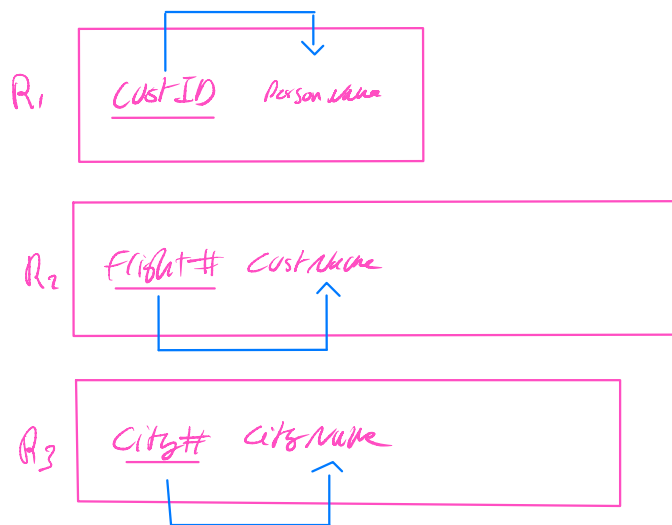
2NF:



3NF:



Final Schema:



كان في كمان جدولين صح وخطأ ثانية واضحة ما فيها صعوبة، وبعض الأسئلة اللي حطبتها تقريبيه لكن نفس الفكرة ١٠٠٪ مجرد اختلاف فالمحتوى، سؤال رسم الاي ار كان طويل شوي ف تدربوا على أسئلة طويلة.

بالنسبة للكويريز كان في اختياري عن الاوت بوت بعد، غير اللي فوق.

GoodLuck <3